$$\lim_{T\to\infty} \frac{1}{T}\int_0^T dt\; e^{-iHt} A\, e^{iHt}$$

H, A linear maps $V \to V$

V → Hilbert space

H has a discrete orthonormal basis of eigenvectors

$\{ |\lambda, a\rangle \mid \lambda \in \Lambda \subseteq \mathbb{R},\ a \in I_\lambda \subseteq \mathbb{N}_0 \}$ basis

$H|\lambda, a\rangle = \lambda |\lambda, a\rangle$

(λ: eigenvalue; a: internal label)

$V_\lambda = \text{span}\{ |\lambda, a\rangle \mid a \in I_\lambda \}$ eigenspace with eigenvalue λ

ex: $H = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$

3 eigenvectors: $v_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ $v_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$

$v_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

label them by eigenvalue:

$\lambda = 1 \Rightarrow v_1, v_2$

$\lambda = 2 \Rightarrow v_3$

$v_1 \equiv |\underset{\lambda}{1}\ \underset{\text{internal ind}}{1}\rangle$ $v_2 = |1\ 2\rangle$ $v_3 = |2\ 1\rangle$

$$A|\lambda a\rangle = \sum_{\lambda' \in \Lambda} \sum_{b \in I_{\lambda'}} A^{\lambda a}_{\lambda' b} |\lambda' b\rangle$$

$\in \mathbb{C}$

action of A on basis

$H|\lambda a\rangle = \lambda |\lambda a\rangle$

$$\overbrace{\lim_{T\to\infty} \frac{1}{T}\int_0^T dt\; e^{-iHt} A\, e^{iHt}}^{\langle A \rangle} |\lambda a\rangle = ?$$

$$e^{iHt}|\lambda a\rangle = \left(\mathbb{1} + iHt + \tfrac{1}{2}(iHt)^2 + \dots\right)|\lambda a\rangle$$

$$= |\lambda a\rangle + i\lambda t |\lambda a\rangle + \tfrac{1}{2}(i\lambda t)^2 |\lambda a\rangle + \dots$$

$$= e^{i\lambda t}|\lambda a\rangle$$

$$e^{-iHt} A\, e^{iHt} |\lambda a\rangle = e^{-iHt} A\, e^{i\lambda t} |\lambda a\rangle$$

$$= \underbrace{e^{i\lambda t}}_{\in \mathbb{C}} e^{-iHt} \sum_{\lambda' \in \Lambda} \sum_{b \in I_{\lambda'}} \overbrace{A^{\lambda a}_{\lambda' b}}^{\in \mathbb{C}} |\lambda' b\rangle$$

$$= e^{i(\lambda-\lambda')t} \sum_{\lambda' \in \Lambda} \sum_{b \in I_{\lambda'}} A^{\lambda a}_{\lambda' b} |\lambda' b\rangle$$

if $\lambda \neq \lambda'$

$$\lim_{T\to\infty} \frac{1}{T} \left. \frac{e^{i(\lambda-\lambda')t}}{i(\lambda-\lambda')} \right|_0^T \to 0$$

$$\langle A\rangle |\lambda a\rangle = \lim_{T\to\infty} \frac{1}{T} \int_0^T dt \; e^{i(\lambda-\lambda')t} \sum_{\lambda' \in \Lambda} \sum_{b \in I_{\lambda'}} A^{\lambda a}_{\lambda' b} |\lambda' b\rangle$$

$$\to \begin{cases} 1 & \text{if } \lambda = \lambda' \\ 0 & \text{if } \lambda \neq \lambda' \end{cases} = \delta_{\lambda\lambda'}$$

$$\langle A\rangle |\lambda a\rangle = \sum_{b \in I_\lambda} A^{\lambda a}_{\lambda b} |\lambda b\rangle$$

① $[\langle A\rangle, H] = 0$ $\quad$ $\langle A\rangle$ preserves eigenspaces : $\langle A\rangle|_{V_\lambda} \subseteq V_\lambda$

$H|_{V_\lambda} = \lambda \mathbb{1}|_{V_\lambda}$

$$\langle A\rangle H |\lambda a\rangle = \sum_{b \in I_\lambda} A^{\lambda a}_{\lambda b} \lambda |\lambda b\rangle$$

$$H\langle A\rangle |\lambda a\rangle = H \sum_{b \in I_\lambda} A^{\lambda a}_{\lambda b} |\lambda b\rangle = \sum_b A^{\lambda a}_{\lambda b} \lambda |\lambda b\rangle$$

$$H = \begin{pmatrix} \boxed{\begin{matrix} \lambda & & \\ & \ddots & \\ & & \lambda \end{matrix}} & & \\ & \boxed{\begin{matrix} \lambda' & \\ & \lambda \end{matrix}} & \\ & & \boxed{} \end{pmatrix}$$

$$A = \begin{pmatrix} \boxed{} & \times & \times \\ \times & \boxed{} & \times \\ \times & \times & \boxed{} \end{pmatrix}$$

② P_λ projection onto V_λ $\qquad$ $P_\lambda = \sum_{b \in I_\lambda} |\lambda b\rangle\langle \lambda b|$

$$\boxed{\langle A\rangle = \sum_\lambda P_\lambda A P_\lambda}$$

$\langle A\rangle$ is "projection" of A onto things that commute with H